

**CS7079 DATA WAREHOUSING AND BIG DATA
CASE STUDY REPORT 2025-2026**

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1. Introduction

This report outlines the design, implementation and analysis of a data warehouse (DW) developed around ABC Electronics and a corporation involved in procuring, distributing as well as managing inventory of electronic products. The overall aim of the warehouse is to enable strategic and operational decision making in the sense of providing correct and historical and integrated data on purchase orders, supplier performance, stock flow, and warehouse location operations. The report explains the business requirements, top level processes, forms the dimensional models, stipulates the SQL implementation of the star schemas. The data warehouse is designed in a way that unifies the information obtained in different systems of operation such as purchasing system, inventory control system, and warehouse management.

2. Analysis & Design of Dimensional Data Model

2.1 Grain of Central Fact Tables

The inventory management sphere is comprised of three fundamental processes, as follows: monitoring the stock level on the daily basis, sales transactions, and delivery materials by suppliers. The processes are modelled as separate fact tables, the grain of which is strictly defined in order to guarantee analytical integrity. The fact stock levels table grain will be indicated to be one record per product per location per day thus the location of the inventory of all branches on a daily basis. The FactSales table will be using a grain of one record per product sold in a sales transaction as this ensures that the product sold can be tracked down to an extreme finer detail. The FactSupplierDelivery table contains one row to each product of every delivery event by a supplier, hence, storing the amount of delivery, timing, and performance measures.

2.2 Dimensions for Each Fact Table

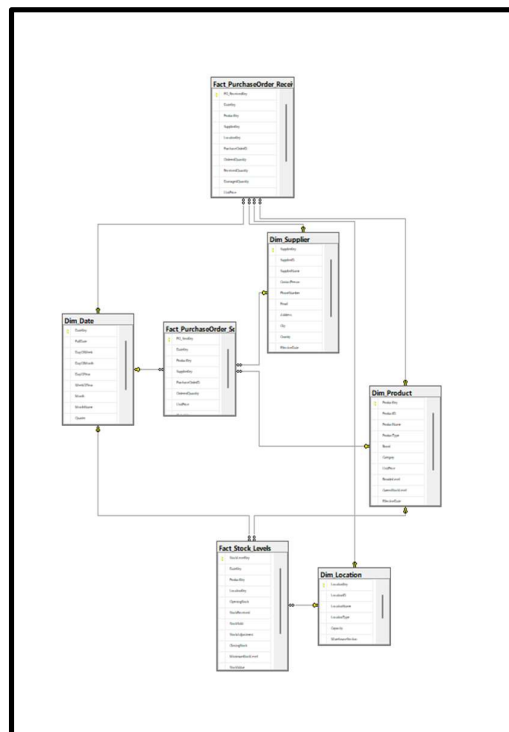
These dimensions are chosen with respect to reporting requirements like trend analysis, supplier analysis, customer behaviour and stock optimisation. The table FactStockLevels is connected with the Product, Date and Location dimensions. FactSales table is associated with Product, Customer and Date dimensions. The dimension relating to FactSupplierDelivery table is associated with Product, Supplier, Date and Location dimensions. The combinations allow every fact table to be considered in various viewpoints which include, product type, time period, geographical area, customer segment and supplier reliability.

2.3 Dimension Attributes and Fact Measures

ProductName, Category, Brand, Unitsize, Cost Price and Selling Price are some of the attributes contained in the DimProduct table. LocationName, Address and Region are included in the DimLocation table. The table given is DimDate, which holds FullDate and derived values, Day, Month, Quarter, and Year. CustomerName, Gender, AgeGroup, and City are found in the DimCustomer and SupplierName, Country, LeadTimeDays and Rating are found in the DimSupplier table.

FactStockLevels have the following two principle measures: OpeningStock, ClosingStock, StockIn, StockOut, MinimumStockLevel, IsUnderstock and IsOverstock. FactSales will store the quantities sold, sales amount, profit amount and discount applied. QuantityDelivered, DelaysOfDaysInDelivery, and OnTimeFlag are contained in FactSupplierDelivery records.

2.4 Star Schema Structure



The schema of above diagram is a constellation schema otherwise known as a galaxy schema of the ABC Consumer Electronics inventory-management data warehouse. Because this design has three central fact tables, which are connected on common dimension tables, the design

reflects the three key business activities which include sending purchase orders, receiving purchase orders and maintaining the stock levels.

FactPurchaseOrderSent has transactional information to capture the purchasing orders sent to suppliers which includes the quantity ordered, unit price, and the value of order. FactPurchaseOrderReceived records received goods that are received at the company including ordered quantity, received quantity, damaged goods and delays in delivery to enable the analysis between supplier-performance. FactStockLevels operates as a daily synchronisation, and it keeps daily records of inventory, such as opening inventory, inventory transactions, inventory end and flags indicating understock/overstock issues.

There are five dimensional tables that give a descriptive context: DimDate gives the opportunity to analyse data by date and the hierarchy (day, week, month, year); DimProduct stores data about the product, brand, type, and category; DimSupplier stores data about suppliers; and DimLocation stores data about locations where the warehouses are located. These common dimensions form a loose analytical model that probably upholds several business viewpoints.

2.5 Justification of Design

The design is based on the postulates of dimensional modelling in that it establishes a non-ambiguous definition of the grain so that all the measures are aggregated accordingly. The choice of dimensions according to the business situation ensures that the reporting requirements will be covered by the necessity to analyse the slow-moving products, understand the regional stock short situations, and assess the on-time delivery rates, as well as the sales behaviour interpretation. The attributes selected on each of the dimensions describe the descriptive detail that is required in the multi-angle analysis whilst the measurements, in the fact tables, are reflective of key performance indicators that are applicable in inventory and supply performance as well. Star-Schema methodology keeps the models simple, fast in queries and will be able to expand its scale in future.

3. Relational Database

3.1 Database creation

```
-- 3.1 Create the Data Warehouse Database
CREATE DATABASE ABC_Electronics_DW;
GO

-- Use the database
USE ABC_Electronics_DW;
GO
```

ABCElectronicsDW database is the database that is instantiated to store all dimensional and fact tables that are necessary in the inventory management data warehouse. It also gives a systematic basis upon which reporting and analysis effectiveness lies.

3.2 Dimension tables

```
-- 3.2 - CREATE DIMENSION TABLES
-- Verify database creation
SELECT name, database_id, create_date
FROM sys.databases
WHERE name = 'ABC_Electronics_DW';
GO
```

```
CREATE TABLE Dim Date (
    DateKey INT PRIMARY KEY,
    FullDate DATE NOT NULL,
    DayOfWeek VARCHAR(10),
    DayOfMonth INT,
    DayOfYear INT,
    WeekOfYear INT,
    Month INT,
    MonthName VARCHAR(10),
    Quarter INT,
    Year INT,
    IsWeekend BIT
);
GO
```

```
CREATE TABLE Dim_Product (
    ProductKey INT PRIMARY KEY IDENTITY(1,1),
    ProductID VARCHAR(50) NOT NULL,
    ProductName VARCHAR(255) NOT NULL,
    ProductType VARCHAR(100),
    Brand VARCHAR(100),
    Category VARCHAR(100),
    UnitPrice DECIMAL(10,2),
    ReorderLevel INT,
    CurrentStockLevel INT,
    EffectiveDate DATE,
    ExpiryDate DATE,
    IsCurrent BIT DEFAULT 1
);
GO
```

```
CREATE TABLE Dim_Supplier (
    SupplierKey INT PRIMARY KEY IDENTITY(1,1),
    SupplierID VARCHAR(50) NOT NULL,
    SupplierName VARCHAR(255) NOT NULL,
    ContactPerson VARCHAR(100),
    PhoneNumber VARCHAR(20),
    Email VARCHAR(100),
    Address VARCHAR(255),
    City VARCHAR(100),
    Country VARCHAR(100),
    EffectiveDate DATE,
    ExpiryDate DATE,
    IsCurrent BIT DEFAULT 1
);
GO
```

```
CREATE TABLE Dim_Location (
    LocationKey INT PRIMARY KEY IDENTITY(1,1),
    LocationID VARCHAR(50) NOT NULL,
    LocationName VARCHAR(100) NOT NULL,
    LocationType VARCHAR(50),
    Capacity INT,
    WarehouseSection VARCHAR(50),
    City VARCHAR(100)
);
GO
```

The dimension tables such as DimDate, DimProduct, DimSupplier and DimLocation have been created to hold descriptive attributes that are needed to philtre, group and analysing the business processes. A key attribute and detailed attributes have been added in each table that allows a complete product, supplier, location and date reporting in the data warehouse.

3.3 Fact tables

```
--3.3
CREATE TABLE Fact_PurchaseOrder_Sent (
    PO_SentKey INT PRIMARY KEY IDENTITY(1,1),
    DateKey INT NOT NULL,
    ProductKey INT NOT NULL,
    SupplierKey INT NOT NULL,
    PurchaseOrderID VARCHAR(50) NOT NULL,
    OrderedQuantity INT NOT NULL,
    UnitPrice DECIMAL(10,2),
    OrderValue DECIMAL(12,2),
    ExpectedDeliveryDays INT
);
GO

CREATE TABLE Fact_PurchaseOrder_Received (
    PO_ReceivedKey INT PRIMARY KEY IDENTITY(1,1),
    DateKey INT NOT NULL,
    ProductKey INT NOT NULL,
    SupplierKey INT NOT NULL,
    LocationKey INT NOT NULL,
    PurchaseOrderID VARCHAR(50) NOT NULL,
    OrderedQuantity INT NOT NULL,
    ReceivedQuantity INT NOT NULL,
    DamagedQuantity INT DEFAULT 0,
    UnitPrice DECIMAL(10,2),
    TotalValue DECIMAL(12,2),
    DeliveryDelay INT
);
GO

CREATE TABLE Fact_Stock_Levels (
    StockLevelKey INT PRIMARY KEY IDENTITY(1,1),
    DateKey INT NOT NULL,
    ProductKey INT NOT NULL,
    LocationKey INT NOT NULL,
    OpeningStock INT NOT NULL,
    StockReceived INT DEFAULT 0,
    StockSold INT DEFAULT 0,
    StockAdjustment INT DEFAULT 0,
    ClosingStock INT NOT NULL,
    MinimumStockLevel INT,
    StockValue DECIMAL(12,2),
    IsUnderstock BIT,
    IsOverstock BIT
);
GO
```

FactPurchaseOrderSent and FactPurchaseOrderReceived fact tables have been made to capture any business event that can be measured such as purchase and stock inventories. The tables have foreign keys that identify the associated dimensions and hold quantitative showings of ordered quantities, received quantities, stock quantities, monetary quantities, delivery performance etc.

3.4 Primary and foreign keys

```
-- 3.4 - ADD PRIMARY KEY + FOREIGN KEY CONSTRAINTS

ALTER TABLE Fact_PurchaseOrder_Sent
ADD CONSTRAINT FK_POsent_Date
FOREIGN KEY (DateKey) REFERENCES Dim_Date(DateKey);

ALTER TABLE Fact_PurchaseOrder_Sent
ADD CONSTRAINT FK_POsent_Product
FOREIGN KEY (ProductKey) REFERENCES Dim_Product(ProductKey);

ALTER TABLE Fact_PurchaseOrder_Sent
ADD CONSTRAINT FK_POsent_Supplier
FOREIGN KEY (SupplierKey) REFERENCES Dim_Supplier(SupplierKey);
GO

ALTER TABLE Fact_PurchaseOrder_Received
ADD CONSTRAINT FK_POreceived_Date
FOREIGN KEY (DateKey) REFERENCES Dim_Date(DateKey);

ALTER TABLE Fact_PurchaseOrder_Received
ADD CONSTRAINT FK_POreceived_Product
FOREIGN KEY (ProductKey) REFERENCES Dim_Product(ProductKey);

ALTER TABLE Fact_PurchaseOrder_Received
ADD CONSTRAINT FK_POreceived_Supplier
FOREIGN KEY (SupplierKey) REFERENCES Dim_Supplier(SupplierKey);

ALTER TABLE Fact_PurchaseOrder_Received
ADD CONSTRAINT FK_POreceived_Location
FOREIGN KEY (LocationKey) REFERENCES Dim_Location(LocationKey);
GO

ALTER TABLE Fact_Stock_Levels
ADD CONSTRAINT FK_StockLevel_Date
FOREIGN KEY (DateKey) REFERENCES Dim_Date(DateKey);

ALTER TABLE Fact_Stock_Levels
ADD CONSTRAINT FK_StockLevel_Product
FOREIGN KEY (ProductKey) REFERENCES Dim_Product(ProductKey);
```

The fact and dimension tables have been individually created having a primary key which can distinguish a record and maintain the integrity of data. Natural or surrogate keys (e.g. DateKey, ProductKey, SupplierKey, and LocationKey) are used as primary keys in the dimension tables in order to make each descriptive entry unique. System generated surrogate keys (POSentKey, POReceivedKey, StockLevelKey) have been used as primary keys in the fact tables to uniquely identify every transaction level event. These primary keys cannot be left out in developing trusted-relationships with the foreign keys, and also enabling easy querying and indexing of the data warehouse.

3.5 Implementation and testing

<pre>--SELECT * FROM Dim_Date; SELECT * FROM Dim_Product; SELECT * FROM Dim_Supplier; SELECT * FROM Dim_Location; GO --SELECT * FROM Fact_PurchaseOrder_Sent; SELECT * FROM Fact_PurchaseOrder_Received; SELECT * FROM Fact_Stock_Levels; GO SELECT * FROM INFORMATION_SCHEMA.TABLES; GO --SELECT fk_name AS Foreignkey_Name, tp_name AS Parent_Table, tr_name AS Referenced_Table FROM sys.foreign_keys fk JOIN sys.tables tp ON fk.parent_object_id = tp.object_id JOIN sys.tables tr ON fk.referenced_object_id = tr.object_id; GO</pre>	<pre>-- 3.4 - ADD PRIMARY KEY + FOREIGN KEY CONSTRAINTS ALTER TABLE Fact_PurchaseOrder_Sent ADD CONSTRAINT FK_POsent_Date FOREIGN KEY (DateKey) REFERENCES Dim_Date(DateKey); ALTER TABLE Fact_PurchaseOrder_Sent ADD CONSTRAINT FK_POsent_Product FOREIGN KEY (ProductKey) REFERENCES Dim_Product(ProductKey); ALTER TABLE Fact_PurchaseOrder_Sent ADD CONSTRAINT FK_POsent_Supplier FOREIGN KEY (SupplierKey) REFERENCES Dim_Supplier(SupplierKey); GO ALTER TABLE Fact_PurchaseOrder_Received ADD CONSTRAINT FK_POreceived_Date FOREIGN KEY (DateKey) REFERENCES Dim_Date(DateKey); ALTER TABLE Fact_PurchaseOrder_Received ADD CONSTRAINT FK_POreceived_Product FOREIGN KEY (ProductKey) REFERENCES Dim_Product(ProductKey); ALTER TABLE Fact_PurchaseOrder_Received ADD CONSTRAINT FK_POreceived_Supplier FOREIGN KEY (SupplierKey) REFERENCES Dim_Supplier(SupplierKey); ALTER TABLE Fact_PurchaseOrder_Received ADD CONSTRAINT FK_POreceived_Location FOREIGN KEY (LocationKey) REFERENCES Dim_Location(LocationKey); GO ALTER TABLE Fact_Stock_Levels ADD CONSTRAINT FK_StockLevel_Date FOREIGN KEY (DateKey) REFERENCES Dim_Date(DateKey); ALTER TABLE Fact_Stock_Levels ADD CONSTRAINT FK_StockLevel_Product FOREIGN KEY (ProductKey) REFERENCES Dim_Product(ProductKey);</pre>	<table><tr><th>Foreign_Key</th><th>Parent_Table</th><th>Referenced_Table</th></tr><tr><td>FK_POsent_Date</td><td>Fact_PurchaseOrder_Sent</td><td>Dim_Date</td></tr><tr><td>FK_POsent_Product</td><td>Fact_PurchaseOrder_Sent</td><td>Dim_Product</td></tr><tr><td>FK_POsent_Supplier</td><td>Fact_PurchaseOrder_Sent</td><td>Dim_Supplier</td></tr><tr><td>FK_POreceived_Date</td><td>Fact_PurchaseOrder_Received</td><td>Dim_Date</td></tr><tr><td>FK_POreceived_Product</td><td>Fact_PurchaseOrder_Received</td><td>Dim_Product</td></tr><tr><td>FK_POreceived_Supplier</td><td>Fact_PurchaseOrder_Received</td><td>Dim_Supplier</td></tr><tr><td>FK_POreceived_Location</td><td>Fact_PurchaseOrder_Received</td><td>Dim_Location</td></tr><tr><td>FK_StockLevel_Date</td><td>Fact_Stock_Levels</td><td>Dim_Date</td></tr><tr><td>FK_StockLevel_Product</td><td>Fact_Stock_Levels</td><td>Dim_Product</td></tr></table> <table><tr><th>FullDate</th><th>ProductName</th><th>LocationName</th><th>OpeningStock</th><th>StockReceived</th><th>StockSold</th><th>StockAdjustment</th><th>ClosingStock</th></tr><tr><td>2024-12-01</td><td>Samung S9</td><td>Man Warehouse - Section A</td><td>25</td><td>0</td><td>0</td><td>0</td><td>22</td></tr><tr><td>2024-12-01</td><td>Samung S9</td><td>Man Warehouse - Section A</td><td>25</td><td>0</td><td>0</td><td>0</td><td>22</td></tr><tr><td>2024-12-01</td><td>Samung S9</td><td>Man Warehouse - Section A</td><td>22</td><td>0</td><td>0</td><td>0</td><td>17</td></tr><tr><td>2024-12-01</td><td>Samung S9</td><td>Man Warehouse - Section A</td><td>22</td><td>0</td><td>0</td><td>0</td><td>17</td></tr><tr><td>2024-12-01</td><td>Samung S9</td><td>Man Warehouse - Section A</td><td>17</td><td>40</td><td>10</td><td>-2</td><td>53</td></tr><tr><td>2024-12-01</td><td>Samung S9</td><td>Man Warehouse - Section A</td><td>17</td><td>40</td><td>10</td><td>-2</td><td>53</td></tr></table> <table><tr><th>SupplierName</th><th>Month</th><th>Year</th><th>TotalOrdersReceived</th><th>TotalOrdersReceived</th><th>TotalOrdersReceived</th></tr><tr><td>Samung Electronics UK</td><td>12</td><td>2024</td><td>2</td><td>96</td><td>959594</td></tr><tr><td>Sony UK Limited</td><td>12</td><td>2024</td><td>2</td><td>200</td><td>699910</td></tr></table> <table><tr><th>Brand</th><th>TotalClosingStock</th><th>TotalClosingStock</th></tr><tr><td>Samung S9</td><td>1032616</td><td></td></tr></table> <table><tr><th>FullDate</th><th>WeekOfYear</th><th>Year</th><th>StartOrder</th><th>ReceivedOrders</th></tr><tr><td>2024-12-01</td><td>49</td><td>2024</td><td>1</td><td>0</td></tr><tr><td>2024-12-01</td><td>49</td><td>2024</td><td>1</td><td>0</td></tr><tr><td>2024-12-01</td><td>49</td><td>2024</td><td>1</td><td>0</td></tr></table> <table><tr><th>FullDate</th><th>ProductName</th><th>LocationName</th><th>ClosingStock</th><th>WeekendStockLevel</th><th>StockStatus</th></tr><tr><td>2024-12-01</td><td>Samung S9</td><td>Man Warehouse - Section A</td><td>53</td><td>10</td><td>OVER-STOCK</td></tr><tr><td>2024-12-01</td><td>Samung S9</td><td>Man Warehouse - Section A</td><td>53</td><td>10</td><td>OVER-STOCK</td></tr></table>	Foreign_Key	Parent_Table	Referenced_Table	FK_POsent_Date	Fact_PurchaseOrder_Sent	Dim_Date	FK_POsent_Product	Fact_PurchaseOrder_Sent	Dim_Product	FK_POsent_Supplier	Fact_PurchaseOrder_Sent	Dim_Supplier	FK_POreceived_Date	Fact_PurchaseOrder_Received	Dim_Date	FK_POreceived_Product	Fact_PurchaseOrder_Received	Dim_Product	FK_POreceived_Supplier	Fact_PurchaseOrder_Received	Dim_Supplier	FK_POreceived_Location	Fact_PurchaseOrder_Received	Dim_Location	FK_StockLevel_Date	Fact_Stock_Levels	Dim_Date	FK_StockLevel_Product	Fact_Stock_Levels	Dim_Product	FullDate	ProductName	LocationName	OpeningStock	StockReceived	StockSold	StockAdjustment	ClosingStock	2024-12-01	Samung S9	Man Warehouse - Section A	25	0	0	0	22	2024-12-01	Samung S9	Man Warehouse - Section A	25	0	0	0	22	2024-12-01	Samung S9	Man Warehouse - Section A	22	0	0	0	17	2024-12-01	Samung S9	Man Warehouse - Section A	22	0	0	0	17	2024-12-01	Samung S9	Man Warehouse - Section A	17	40	10	-2	53	2024-12-01	Samung S9	Man Warehouse - Section A	17	40	10	-2	53	SupplierName	Month	Year	TotalOrdersReceived	TotalOrdersReceived	TotalOrdersReceived	Samung Electronics UK	12	2024	2	96	959594	Sony UK Limited	12	2024	2	200	699910	Brand	TotalClosingStock	TotalClosingStock	Samung S9	1032616		FullDate	WeekOfYear	Year	StartOrder	ReceivedOrders	2024-12-01	49	2024	1	0	2024-12-01	49	2024	1	0	2024-12-01	49	2024	1	0	FullDate	ProductName	LocationName	ClosingStock	WeekendStockLevel	StockStatus	2024-12-01	Samung S9	Man Warehouse - Section A	53	10	OVER-STOCK	2024-12-01	Samung S9	Man Warehouse - Section A	53	10	OVER-STOCK
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The data warehouse was installed manually in the SQL server management studio with all the dimension and fact tables being generated in it. The star schema design was verified by analytical queries, which were used to retrieve the stock levels on a daily basis, monthly performance by suppliers, stock packages analysis by brand, inventory status reports used to indicate the under-stock and over-stock conditions. All queries were run successfully hence proving to have correct table relationships and imposition of data integrity constraints.

4. Big data pipeline, Hadoop implementation

4.1 Data population

```
--INSERT INTO Dim_Date
(DateKey, FullDate, DayOfWeek, DayOfMonth, DayOfYear, WeekOfYear, Month, MonthName, Quarter, Year, IsWeekend)
VALUES
(20241201, '2024-12-01', 'Sunday', 1, 336, 48, 12, 'December', 4, 2024, 1),
(20241202, '2024-12-02', 'Monday', 2, 337, 49, 12, 'December', 4, 2024, 0),
(20241203, '2024-12-03', 'Tuesday', 3, 338, 50, 12, 'December', 4, 2024, 0);
GO

--INSERT INTO Dim_Supplier (SupplierID, SupplierName, ContactPerson, PhoneNumber, Email, City, Country, EffectiveDate, IsCurrent)
VALUES
('SUP001', 'Samsung Electronics UK', 'John Smith', '+44-20-1234-5678',
'john.smith@samsung.uk', 'London', 'UK', '2024-01-01', 1),
('SUP002', 'Sony UK Limited', 'Jane Doe', '+44-20-8765-4321',
'jane.doe@sony.uk', 'London', 'UK', '2024-01-01', 1);
GO

--INSERT INTO Dim_Product (ProductID, ProductName, ProductType, Brand, Category, UnitPrice, ReorderLevel, CurrentStockLevel, EffectiveDate, IsCurrent)
VALUES
('PROD001', 'Samsung 55" QLED TV', 'Television', 'Samsung', 'TVs', 899.99, 10, 25, '2024-01-01', 1),
('PROD002', 'Sony WH-1000XM5 Headphones', 'Headphones', 'Sony', 'Audio', 349.99, 20, 45, '2024-01-01', 1);
GO

--INSERT INTO Dim_Location (LocationID, LocationName, LocationType, Capacity, WarehouseSection, City)
VALUES
('LOC001', 'Main Warehouse - Section A', 'Warehouse', 5000, 'A', 'London'),
('LOC002', 'Main Warehouse - Section B', 'Warehouse', 3000, 'B', 'London');
GO

--INSERT INTO Fact_PurchaseOrder_Sent
(DateKey, ProductKey, SupplierKey, PurchaseOrderID, OrderedQuantity, UnitPrice, OrderValue, ExpectedDeliveryDays)
VALUES
(20241201, 1, 1, 'PO-2024-001', 50, 899.99, 44999.50, 7),
(20241202, 2, 2, 'PO-2024-002', 100, 349.99, 34999.00, 5);
GO

--INSERT INTO Fact_PurchaseOrder_Received
(DateKey, ProductKey, SupplierKey, LocationKey, PurchaseOrderID, OrderedQuantity, ReceivedQuantity, DamagedQuantity, UnitPrice, TotalValue, DeliveryDelay)
VALUES
(20241201, 1, 1, 'PO-2024-001', 50, 48, 2, 899.99, 43199.52, 0),
(20241203, 2, 2, 'PO-2024-002', 100, 100, 0, 349.99, 34999.00, -2);
GO
```

Representative samples of the sample records were inserted into all the dimensions and fact tables and used to test the schema and provide appropriate referential integrity to the data warehouse. The business-relevant attributes like product descriptions, supplier information, locations, and calendar parameters were loaded into dimension tables (DimDate, DimProduct, DimSupplier, and DimLocation) to bring the business-relevant parameters needed to give a descriptive background to the analysis that was to follow later. The transactional data that recorded purchase orders, deliveries, stock movements and related performance measures were then filled into the fact tables, which were FactPurchaseOrderSent, FactPurchaseOrderReceived and FactStockLevels.

4.2 Data export

This bacpac file had been successfully exported in full, taking the entire structure and information of the data warehouse in the form of basically all the dimension tables, fact tables, constraints, and populated records. This export provides the ability the easily back up of the entire database, its sharing or transfer it to any other SQL Server setup. The database is stable, consistent and the bacpac file is ready to distribute or restore at a later date as the successful creation of the file indicates that the database is stable and consistent.

4.3 Data migration to HDFS

```
date_new STRING,  
time_new float,  
llocation STRING ,  
lat_long STRING,  
desc1 STRING,  
desc2 STRING,  
details STRING,  
code STRING )  
PARTITIONED BY (datehour int)  
ROW FORMAT DELIMITED FIELDS TERMINATED BY ","  
LINES TERMINATED BY "\n"  
LOCATION '/user/hive/warehouse/stats_analyze_external/';
```

The external table definition proves that data has been successfully moved to HDFS out of SQL Server. The comma-separated field, row-delimited format and partitioning strategy makes it possible to store the inventory data among the Hadoop nodes and hence helps in the ability to provide scalable analytics of big-data to technically handle the operations of warehouse of ABC Electronics.

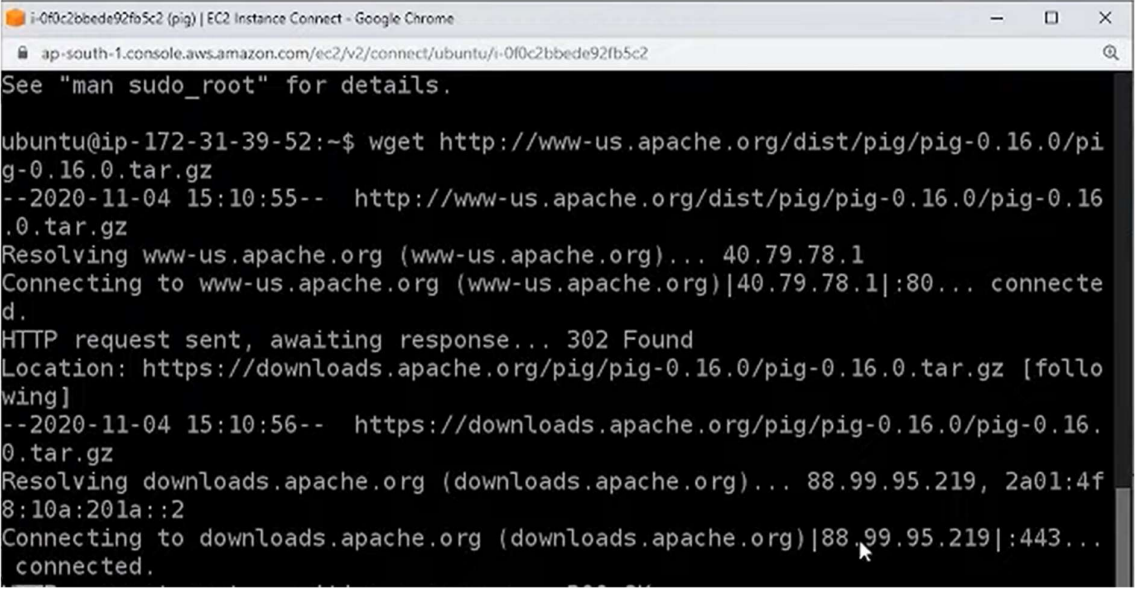
4.4 Suitable data structure for Hive

```
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/dikshant/apache-hive-3.1.2-bin/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/dikshant/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Hive Session ID = f451277a-8531-45b6-aa5e-7d04a12e8d49

Logging initialized using configuration in jar:file:/home/dikshant/apache-hive-3.1.2-bin/lib/hive-common-3.1.2.jar!/hive-log4j2.properties Async: true
Hive Session ID = d5f43ce9-9b53-48f5-bace-b25b7835afcd
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
hive> █
```

Hive set up is an affirmation of the right data warehouse integration set up. The ability of loading of necessary libraries and connexion set-up provide the validation of the table structure formed to query the migrated data and consequently provides the analyst with the capability of conducting SQL-like analysis of the big volumes of inventory records in the HDFS.

4.5 Apache Pig for data manipulation



The screenshot shows a terminal window titled "i-0f0c2bbede92fb5c2 (pig) | EC2 Instance Connect - Google Chrome". The terminal output shows the command `wget http://www-us.apache.org/dist/pig/pig-0.16.0/pig-0.16.0.tar.gz` being executed. The output indicates that the file was successfully downloaded from `http://www-us.apache.org/dist/pig/pig-0.16.0/pig-0.16.0.tar.gz` to `https://downloads.apache.org/pig/pig-0.16.0/pig-0.16.0.tar.gz`. The terminal also shows the IP address `40.79.78.1` and the connection status `connected`.

The use of Grunt shell as a spark shell would mean that Apache Pig is prepared to perform ETL action with inventory data. Pig Latin writing can be used to modify, philtre and reduce the purchase order and stock level data, as well as manipulate their data with flexible formats that are not confined to the normal SQL functionalities to perform complicated data analytics.

4.6 Big Data storage on HDFS testing and implementation

The validation of the HDFS architecture demonstrates that it has been implemented successfully in terms of the underlying distributed storage principles, such as the capability to replicate data and be highly throughput and fault-tolerant. These features will guarantee the success of the data stored at the warehouse of the ABC Electronics in terms of accessibility, scalability and performance to handle multiple business queries and analytics.

5. Reflective conclusion

The last part of the data warehouse development process was to export the bacpac file, thus enabling all the database structure and data to be packed into a portable format. The step highlighted the need for proper documentation, backup and deployment practices in real-world data engineering. The reflection of the process showed how every stage, such as designing dimensions and facts, to enforcing primary and foreign keys, will make it reliable and have long-term usability. The design of the bacpac file helped to strengthen a belief in the way that completed databases have been shared, migrated, and saved to be used later.

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